

Convertidor Fly-Back

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[Signature]

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$

1) $V_s \text{ nominal} = 300V$ $D = 0,5$ $V_o = 5V$

$$5V = \frac{N_2}{N_1} \frac{0,5}{(1-0,5)} 300V$$

$$\frac{N_2}{N_1} = \frac{1}{60}$$

$$\frac{5V}{300V} = \frac{N_2}{N_1} = \frac{1}{60}$$

2) $\frac{N_2}{N_1} \text{ calculado} = \frac{1}{60}$ $V_{s\text{min}} = 270V$ $V_{s\text{max}} = 330V$

$D_{\text{max}}: (V_{s\text{min}})$

$D_{\text{min}}: (V_{s\text{max}})$

$$5V = \frac{1}{60} \cdot \frac{D}{(1-D)} \cdot 270V$$

$$5V = \frac{1}{60} \cdot \frac{D}{(1-D)} \cdot 330V$$

$$\frac{5V \cdot 60}{270V} = \frac{D}{(1-D)}$$

$$\frac{5V \cdot 60}{330V} = \frac{D}{(1-D)}$$

$$\frac{10}{9} = \frac{D}{(1-D)}$$

$$\frac{10}{11} = \frac{D}{(1-D)}$$

$$10 - 10D = 9D$$

$$10 - 10D = 11D$$

$$10 = 19D$$

$$10 = 21D$$

$$D_{\text{max}} = \frac{10}{19} \approx 0,526$$

$$D_{\text{min}} = \frac{10}{21} \approx 0,476$$

3) $P_s = P_o$

$$D I_{L\text{a}} = \frac{V_o I_R}{V_s}$$

$$270V \leq V_s \leq 330V$$

$$V_s I_s = V_o I_R$$

$$I_{L\text{a}} = \frac{V_o I_R}{V_s D}$$

$$0,5A \leq I_R \leq 3A$$

$$I_s = \frac{V_o I_R}{V_s}$$

$I_{L\text{a max}}: (I_{R\text{max}}, V_{s\text{min}})$ $I_{L\text{a min}}: (I_{R\text{min}}, V_{s\text{max}})$

$$I_s = D I_{L\text{a}}$$

$$I_{L\text{a max}} = \frac{V_o I_{R\text{max}}}{V_{s\text{min}} D_{\text{max}}}$$

$$I_{L\text{a min}} = \frac{V_o I_{R\text{min}}}{V_{s\text{max}} D_{\text{min}}}$$

$$I_{L\text{a max}} = \frac{5V \cdot 3A}{270V \cdot \frac{10}{19}}$$

$$I_{L\text{a min}} = \frac{5V \cdot 0,5A}{330V \cdot \frac{10}{21}}$$

$$I_{L\text{a max}} = \frac{19}{180} A = 105,5 \text{ mA}$$

$$I_{L\text{a min}} = \frac{7}{440} A = 15,90 \text{ mA}$$

$$4) \textcircled{I} I_s = D I_{Lm} \quad (\text{Viendo a } D \text{ sea } \text{máx o mín y } I_{Lm} \text{ máx o mín})$$

$$I_{s \text{ máx}}: (D_{\text{máx}}, I_{Lm \text{ máx}})$$

$$I_{s \text{ mín}}: (D_{\text{mín}}, I_{Lm \text{ mín}})$$

$$I_{s \text{ máx}} = D_{\text{máx}} \cdot I_{Lm \text{ máx}}$$

$$I_{s \text{ máx}} = \frac{10}{19} \cdot \frac{19}{180} \text{ A} = \frac{1}{18} \text{ A} = 55,5 \text{ mA}$$

$$I_{s \text{ mín}} = D_{\text{mín}} \cdot I_{Lm \text{ mín}}$$

$$I_{s \text{ mín}} = \frac{10}{21} \cdot \frac{7}{440} \text{ A} = \frac{1}{132} \text{ A} = 7,57 \text{ mA}$$

II

$$I_s = \frac{V_o I_R}{V_s}$$

$$I_{s \text{ máx}}: (I_{R \text{ máx}}, V_{s \text{ mín}})$$

$$I_{s \text{ mín}}: (I_{R \text{ mín}}, V_{s \text{ máx}})$$

$$I_{s \text{ máx}} = \frac{V_o I_{R \text{ máx}}}{V_{s \text{ mín}}}$$

$$I_{s \text{ mín}} = \frac{V_o I_{R \text{ mín}}}{V_{s \text{ máx}}}$$

$$I_{s \text{ máx}} = \frac{5 \text{ V} \cdot 3 \text{ A}}{270 \text{ V}} = \frac{1}{18} \text{ A}$$

$$I_{s \text{ mín}} = \frac{5 \text{ V} \cdot 0,9 \text{ A}}{330 \text{ V}} = \frac{1}{132} \text{ A}$$

$$5) I_{Lm \text{ mín}} = \frac{7}{440} \text{ A} = 15,9 \text{ A} \quad ; \text{ o esta corriente se opera en MCC}$$

Por la forma de onda dada en un MCC, el valor máximo que toma la corriente es del doble que lo mínimo. Por esta razón el valor mínimo que se toma con esta corriente mínima es:

$$I_{Lm \text{ mín máx}} = 2 \cdot I_{Lm \text{ mín}} = 2 \cdot \frac{7}{440} \text{ A} = \frac{7}{220} \text{ A} = 31,81 \text{ mA}$$

$$\text{y como } I_o = I_{Lm} \cdot \left(\frac{N_1}{N_2} \right)$$

$$\frac{N_2}{N_1} = \frac{1}{60} \Rightarrow \frac{N_1}{N_2} = 60$$

$$I_{o \text{ máx}} = I_{Lm \text{ máx}} \cdot \left(\frac{N_1}{N_2} \right)$$

$$I_{o \text{ máx}} = \frac{7}{220} \text{ A} \cdot 60 = \frac{21}{11} \text{ A} = 1,90 \text{ A}$$

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